Product Description

Product Information	
Product name	Active Splitter
Product number	ASP-015-50-A
Manufacturer	FLIM LABS S.r.l.
Registered Address	Via della Farnesina, 3 – 00135 Rome (RM), Italy



General Description

The Active Splitter is a compact electronic module designed for the controlled duplication and distribution of high-frequency electronic signals. It is primarily used in scientific instrumentation, timing systems, event counting, and fluorescence lifetime imaging microscopy (FLIM) applications, where reliable signal replication without degradation of timing or electrical characteristics is required.

The device is housed in a compact aluminum enclosure suitable for benchtop laboratory use or integration into portable measurement systems. The enclosure provides mechanical robustness and electromagnetic shielding within the specified operating limits. Power is supplied via USB Type-C at 5 V DC.

The Active Splitter features two bidirectional SMA connectors operating in loop-through configuration and one dedicated SMA output. The input signal can be applied to either of the loop-through ports and is forwarded unchanged to the second port. In parallel, the signal is actively replicated and provided at the dedicated output as an impedance-matched copy.

This architecture enables signal distribution to multiple instruments while maintaining signal integrity and proper impedance matching. The device is intended for immediate integration into measurement chains and does not require user configuration.

Intended Use and Misuse

Intended Use

The Active Splitter is intended as a compact electronic module for the distribution and duplication of high-frequency signals by professional users in research laboratories, companies, and universities. It may be used as a stand-alone unit or integrated into custom measurement systems and instrumentation.

Reasonably Foreseeable Misuse

The following uses are not permitted:

- Use outside described applications
- Use with incompatible components
- Operation outside specified limits
- Unauthorized modification
- Use with incorrect power supply

Environmental and Operating Conditions

The device is designed for indoor laboratory use only. Operate the device only within the specified environmental limits:

- Operating temperature: 0°C to +40°C
- Storage temperature: −20°C to +70°C
- Relative humidity: up to 80% non-condensing
- Stable, vibration-free surface

Do not operate the device in explosive atmospheres or in the presence of flammable gases.

Ensure adequate ambient lighting during installation and adjustment to avoid connection errors.